Carbon-Aware Scheduling for Phase-Disaggregated LLM Inference in Cloud and Serverless Platforms: Review and Simulation

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***Abstract*—Interactive large language model (LLM) inference separates into compute-bound prefill and memory-bound decode stages across multi-region cloud clusters and bursty serverless GPU pools. A PRISMA-2020-guided review of 61 works shows that while recent systems address subsets in isolation (including single-cluster disaggregated GreenLLM and colocated multi-datacenter carbon routers), no implemented system jointly integrates prefill/decode disaggregation, wide-area key-value (KV) transfer gating, host and GPU embodied carbon amortization, and serverless checkpoint keep-alive, and 75.4% of reviewed studies omit manufacturing emissions. To evaluate this design space, I formalize an ISO/IEC 21031:2024 Software Carbon Intensity (SCI) control architecture in a trace-driven SimPy 4.1.2 discrete-event simulator validated against published serving and loader baselines. Over 10 paired seeds driven by Azure LLM traces, UK NESO, and Japan TSO grid telemetry, hybrid routing lowers operational SCI by up to 60.8% and total SCI by up to 54.9% on conversational traffic at 0.81%–0.82% SLO violations, and reduces bursty-code SLO violations by 85.2%–86.0% below unconditional geographic disaggregation. On wind-clean grids, amortized hardware reaches 93.0%–97.6% of total SCI, causing four spare clean nodes to raise lifecycle emissions by 27.7%–32.6%.**

***Index Terms*—Carbon-aware scheduling, embodied carbon, large language model inference, phase disaggregation, serverless GPUs, Software Carbon Intensity.**

I. Introduction

LARGE language model (LLM) inference has shifted cloud datacenter energy demand from offline batch training toward continuous, latency-critical serving. Survey estimates indicate that U.S. datacenter electricity consumption could climb from 176 TW h in 2023 toward 580 TW h by 2028 [1]. Because interactive LLM serving runs across distributed cloud clusters, its net carbon footprint depends on compute power draw, temporal and regional electricity grid carbon intensity (gCO_2eq/kWh), and hardware manufacturing emissions amortized over server and accelerator lifetimes [2], [3].

Two distinct research communities have tackled pieces of this problem over the past five years, yet their mechanisms remain only partially connected. First, systems researchers have shown that autoregressive LLM generation divides into a compute-bound *prefill* phase (governing time-to-first-token, TTFT) and a memory-bandwidth-bound *decode* phase (governing time-per-output-token, TPOT) [4]–[6]. Disaggregating prefill and decode across separate GPU pools eliminates phase interference and enables power capping or lower-tier decode GPUs [4], [7]–[9]. Second, sustainable cloud researchers have built carbon-aware spatial/temporal routers and serverless keep-alive controllers that follow low-carbon regions or clean grid hours [10]–[14].

A PRISMA-2020-guided literature review of 61 retained works (47 empirical/modeling studies, 11 surveys or research agendas, 2 standards, and 1 production trace release) clarifies both real progress since 2024 and the remaining architectural gap. On the single-cluster side, GreenLLM [15] pairs heterogeneous prefill/decode disaggregation with grid carbon intensity and GPU embodied lifetime reuse, while SPROUT [16] and GreenCache [17] optimize single-datacenter operational plus hardware/SSD embodied carbon. On the multi-region side, recent geo-distributed LLM routers [18]–[21] achieve substantial operational emission reductions across colocated regional pools. However, across the full 61-work corpus, 46/61 works (75.4%) still omit manufacturing emissions, 0/8 cross-region carbon schedulers implement prefill/decode disaggregation or wide-area key-value (KV) cache transfer economics, and Sisodia's `llm-d` control-plane proposal [22] includes zero empirical evaluations. Crucially, no implemented system in the reviewed literature jointly integrates phase-disaggregated P/D serving, cross-region carbon routing with explicit WAN KV transfer-vs-recompute gating, amortized host and GPU embodied accounting, and hazard-driven serverless GPU checkpoint keep-alive across cold-start loader tiers (multi-gigabyte 70B reload times of 10.3 s to 84.0 s [23]).

To evaluate this joint intersection under controlled, reproducible conditions, I conduct a PRISMA-2020-guided review paired with a trace-driven discrete-event simulation study structured around four research questions:

- **RQ1 (Taxonomy and Joint-Design Gaps):** Where do existing carbon-aware cloud schedulers, geo-distributed LLM routers, serverless keep-alive controllers, and phase-disaggregated serving engines intersect, and what remains unimplemented at their boundary?
- **RQ2 (Disaggregated Carbon–SLO Frontier):** Under real UK/Japan grid telemetry and production Azure LLM traces, when does hybrid carbon-and-queue-aware phase routing outperform latency-only colocation and phase-agnostic or unconditional spatial routing under TTFT/TPOT SLOs?
- **RQ3 (Serverless Keep-Alive Economics):** Across carbon-intensive and wind-clean grids, how does hazard-driven dynamic keep-alive trade operational carbon against cold-start SLO violations as loader latency spans fast NVMe streaming to standard deserialization?

Author identities, affiliations, contact details, ORCID identifiers, and public artifact URLs are omitted for double-blind review.

- **RQ4 (Embodied Scale-Out Paradox):** When does adding spare GPU nodes in clean regions invert net lifecycle Software Carbon Intensity (SCI_{tot}), and how do decoupled host/GPU lifetimes (9 yr/3 yr vs. 4 yr) alter provisioning decisions?

This paper makes three contributions:

1) **PRISMA-2020-Guided Literature Synthesis:** Screening 98 records under explicit single-author plus AI-assisted review disclosure yields a verified 61-work corpus (56 full-text, 5 abstract-only), a dual-axis taxonomy (`L1.A-L1.F`), a direct feature comparison against the closest 2024–2026 systems, and seven open challenges (`OC1-OC7`).
2) **Validated Carbon-Extended Simulator:** I formalize an ISO/IEC 21031:2024 [3] lifecycle model inside SimPy 4.1.2 integrating Splitwise power/step curves [4], GreenSKU/LLMCarbon host-plus-GPU embodied parameters [24], [25], WAN KV transfer-vs-recompute gating, and GapHazard serverless control [14], [23], validated via 24 benchmark checks against published cluster and loader data.
3) **Multi-Seed Trace-Driven Empirical Evaluation:** Across 10 paired seeds using CC-BY-4.0 Azure traces [26], UK NESO, and Japan TSO telemetry (with 24h trace-tiling sensitivity ablation), hybrid routing (`P4`), hazard keep-alive (`S3`), and elastic over-provisioning (`H4`) quantify the joint carbon–SLO–embodied trade-offs detailed in Section VI.

The remainder of the paper is organized as follows. Section II summarizes LLM phases, WAN KV bounds, serverless cold starts, and ISO/IEC 21031:2024 accounting. Section III documents the PRISMA-2020-guided search, screening, and validity threats. Section IV answers `RQ1` and contrasts the closest related systems. Section V details simulator equations, traces, validation benchmarks, and tiling ablation. Section VI presents multi-seed findings (`E1`, `E2`, `SW1-SW4`, `H4`) with paired effect sizes and answers `RQ1-RQ4`. Section VII discusses open challenges and limitations, and Section VIII concludes with Generative AI and artifact availability disclosures.

II. Background and Preliminaries

Building carbon-aware LLM serving schedulers requires joining four domains that developed under separate abstractions: phase-level transformer execution, wide-area carbon routing, GPU serverless cold-start management, and lifecycle carbon accounting under ISO/IEC 21031:2024. Fig. 1 illustrates how these layers interact inside the dual-track control architecture.

A. Autoregressive LLM Phases and Disaggregation

Within cloud platforms [27], autoregressive serving separates into compute-bound *prefill* (parallel prompt processing over L_{in} tokens that populates the KV cache and governs TTFT) and memory-bandwidth-bound *decode* (sequential generation over L_{out} steps reloading weights and KV state from HBM, governing TPOT). Production Azure traces reveal sharp workload skew: conversational traffic averages $1{,}154.7$ input and 211.1 output tokens with steady arrivals (CV $= 1.094$), whereas coding completions average $2{,}047.8$ input and 27.9 output tokens under extreme burstiness (CV $= 13.152$) [26]; Mooncake reaches $8{,}590$ input vs. 182 output tokens [28].

Local schedulers target intra-node bottlenecks first: PagedAttention cuts KV waste from 61.8%–79.6% to 3.7% (saving up to 55.2% KV memory) [29], while Sarathi-Serve eliminates $28.3\times$ stalls via chunked prefill [6]. Phase-disaggregated engines (Splitwise [4], DistServe [5], Mooncake [28], DualScale [8]) decouple prefill and decode pools over NVLink or RDMA. Splitwise shows decode GPUs sustain 50% TDP caps (700 W to 350 W on H100) with minimal TPOT loss at $2.35\times$ iso-power throughput [4], and DualScale saves 39% prefill and 48% decode energy at P99 parity [8]. Decode still dominates energy across token ratios [30], bilinear $(L_{\text{in}}, L_{\text{out}}, L_{\text{in}}L_{\text{out}})$ regression reaches $R^2 \geq 0.964$ [31], and Li et al. [32] caution that small-batch disaggregation can trail colocation until batch-32 KV pressure triggers colocation collapse.

B. Wide-Area Carbon Routing and KV Transfer Economics

Across 123 Electricity Maps zones, Sukprasert et al. [12] measure a $40\times$ grid CI spread (mean 368.39 gCO_2eq/kWh), shrinking 96% unconstrained carbon gains to 51.5% at 50% spare capacity and 31% under $50\,\text{ms}$ RTT limits, with 10%–12% error overhead at 50% forecast noise. Geo-distributed routers (GreenCourier [33] at 8.7%–17.8% carbon cuts, Wait Awhile [10], CarbonScaler [11], EcoVisor [34], Caribou [13], Calavaro et al. [35]) assume deferrable jobs or small HTTP payloads. Cross-region P/D instead ships hundreds of megabytes of 70B FP16 KV state per request, incurring serial WAN delay ($L_{\text{kv}}/B_{\text{wan}}$) and network energy (0.001–0.005 kWh/GB [13]) that can offset grid arbitrage.

C. Serverless GPU Keep-Alive and Checkpoint Loading

With 81% of Azure Functions apps receiving $\leq$ 1 req/min [36], idle nodes draw 21%–83% CPU power [14] and 60–100 W per idle GPU, whereas reloading 70B (130 GB) weights spans 84.0 s (PyTorch), 48.0 s (Safetensors), and 10.3 s (ServerlessLLM RAID-0 [23]). Adaptive controllers (EcoLife [37], CASA [38], LowCarb [39], LACE-RL at -77.08% idle carbon [14], Festina at -56% energy and -85% cold starts [40]) omit cross-grid loader sensitivity.

D. ISO/IEC 21031:2024 SCI and Embodied Carbon

To prevent shifting emissions between operational and hardware budgets, the Green Software Foundation Software Carbon Intensity (SCI) specification, standardized as ISO/IEC 21031:2024 [3], mandates an additive lifecycle intensity per functional unit R (here, 1,000 processed tokens) while explicitly excluding offset purchases:

$$\text{SCI} = \frac{(E \cdot I) + M}{R}, \tag{1}$$

where E is total consumed electricity (kWh, inclusive of datacenter PUE), I is regional grid carbon intensity (gCO_2eq/kWh), and $M = M_{\text{server}} \cdot (t_{\text{slot}}/T_{\text{life}})$ is amortized manufacturing carbon.

Industry GHG audits [2], architectural models [41], [42], and LLM carbon tools [24], [25] show that embodied emissions (800 $kgCO_2eq$ host baseline [24]; 921.6 $kgCO_2eq$ in the GreenSKU Genoa assembly, plus die/HBM [25]) represent

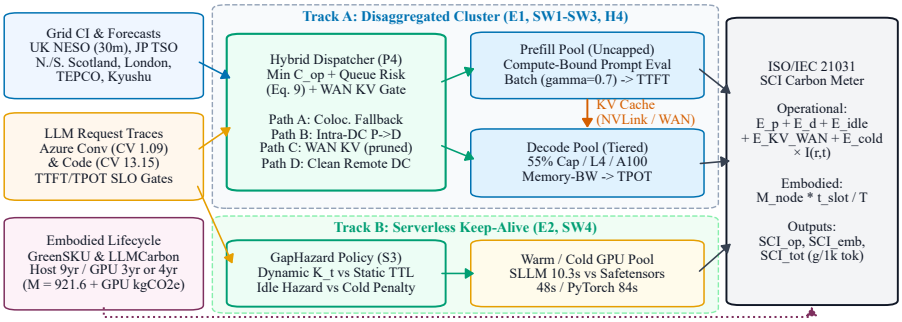


Fig. 1. Dual-track carbon-aware phase-disaggregated LLM cluster (Track A: Paths A–D via Eq. (9)) and serverless GPU keep-alive (Track B: GapHazard policy `S3`) feeding the unified ISO/IEC 21031:2024 SCI telemetry meter.

24%–35% of LLM totals on average grids and surpass 80% as grids decarbonize. Consequently, metrics using raw GPU utilization [43] or ignoring power capping [7], [44], [45] and aging amortization [46], [47] risk optimizing local Watts while raising system-wide lifecycle carbon. These tradeoffs motivate unified simulation testbeds in the tradition of CloudSim [48].

III. PRISMA-2020-Guided Review Methodology

To ground `RQ1` and parameterize the simulator from primary artifacts, I conducted a PRISMA-2020-guided literature review across Crossref, Semantic Scholar, OpenAlex, arXiv, ACM Digital Library, IEEE Xplore, USENIX, and official standards indices.

A. Search Queries, Inclusion Criteria, and Screening

Candidate studies published between 2011 and November 2026 were retrieved across three channels, expanded in revision v2 with targeted searches over 2024–November 2026 preprints and proceedings:

- **Channel 1 (Carbon-Aware Cloud & Serverless Scheduling, 16 records):** Paired carbon-intensity, grid emissions, SCI, and embodied terms with spatial/temporal shifting, serverless keep-alive, and cold-start mitigation (`Query A`).
- **Channel 2 (Phase-Disaggregated LLM Serving & v2 Update, 44 records):** Combined LLM inference, KV caching, prefill/decode disaggregation, and GPU serving with energy, DVFS, power capping, and embodied/SCI queries (`Query B`, plus 2024–2026 arXiv/Crossref follow-ups on `"embodied carbon" AND "LLM" AND "inference"`, `"serverless" AND "keep-alive" AND "LLM" AND "carbon"`, `"Software Carbon Intensity" AND "LLM"`, and `"carbon-aware" + "prefill decode disaggregation"`). Exact boolean search strings and log dates appear in the online supplement.
- **Channel 3 (Surveys, Traces, Lifecycle Models & Snowballing, 38 records):** Aggregated recent surveys, standards, hardware embodied datasets, citation chasing, and 9 systems verified during revision v2 [15], [17]–[22], [49], [50].

Single-Author and AI-Assisted Screening Protocol: All candidate retrieval, deduplication, and full-text eligibility checks were performed by a single human author assisted by generative AI search and extraction tooling; every included entry's bibliographic metadata, DOI/arXiv identifier, and quantitative metrics were verified directly against primary PDFs, HTML preprints, or Crossref records. As shown in Fig. 2, removing 3 duplicates from 98 identified records left 95 unique records for title/abstract screening. Excluding 14 out-of-scope records (lacking cloud serving, datacenter energy, or carbon context, including 3 edge/mobile/satellite works) yielded 81 candidates for eligibility assessment, where 20 more were excluded (offline training only, latency-only disaggregation/pruning, or superseded position notes), leaving 61 included works.

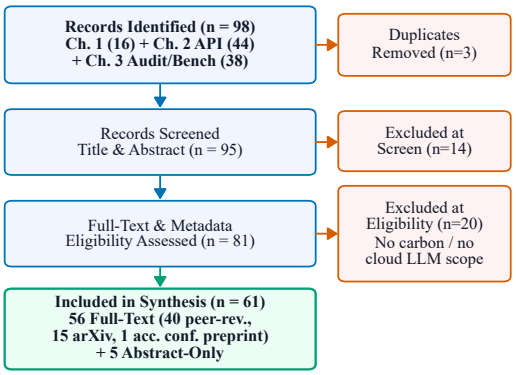


Fig. 2. PRISMA-2020-guided search, screening, and eligibility pipeline yielding the 61-work corpus (56 full-text, 5 abstract-only).

B. Corpus Provenance and Preprint Disclosure

The 61-work corpus comprises 47 empirical/modeling systems, 11 surveys or research agendas, 2 standards (ISO/IEC 21031:2024 [3] and NIST SP 800-145 [27]), and 1 trace release [26]. By recency, 56/61 works (91.8%) appeared in 2022–2026, and 34/61 (55.7%) appeared in 2025–2026. Retrieval depth breaks down as:

1) **Full-Text Audited (56/61, 91.8%):** Complete texts underwent equation and table verification, spanning 40 archival publications/standards/traces plus 16 preprints (15 arXiv-only preprints [1], [8], [9], [15], [18], [20]–[22], [24], [49], [51]–[55] and 1 accepted SIGOPS ATC 2026 preprint [40]).
2) **Abstract-Only Verified (5/61, 8.2%):** Five paywalled records relied strictly on verified abstracts [35], [39], [41], [43], [48], with zero constants imported into the simulator.

IV. Taxonomy and Literature Synthesis

This section answers RQ1 by classifying the 61 retained works into six functional classes (L1.A–L1.F), mapping the 50 non-survey works across phase awareness and carbon signals, and contrasting the closest 2024–2026 systems directly against this study.

A. Dual-Axis Taxonomy (L1.A–L1.F)

1) L1.A Prior Surveys and Research Agendas ($n = 11$): Existing surveys address sustainable cloud scheduling or efficient LLM inference separately (Table I). Hewage et al. [51], Yang et al. [1], Alam et al. [56] (finding 70%, 44/63, of surveyed cloud carbon works rely solely on simulation and 6.3% track live grid CI), Agarwal et al. [57], and Besozzi et al. [58] synthesize cloud and serverless scheduling without P/D LLM serving. Conversely, Miao et al. [59], Jiang et al. [60], Pan et al. [52], Bai et al. [61], and Zhen et al. [54] examine LLM inference disaggregation, KV compression, and cold starts while omitting cross-region carbon arbitrage and embodied emissions. Most closely, Sisodia [22] synthesizes distributed serving and proposes a Kubernetes `llm-d` Sustainable Inference Control Plane (SICP) combining operational, embodied, and intra-cluster KV recompute terms, but explicitly reports zero experiments.

2) L1.B Spatial and Temporal Cloud Carbon Schedulers ($n = 12$): Single-DC temporal schedulers (Wait Awhile [10], CarbonScaler [11], EcoVisor [34], Clover [47]) and generic WAN routers (GreenCourier [33], Caribou [13], Calavaro [35], Sukprasert [12]) establish spatial carbon bounds for non-LLM tasks. Four recent geo-distributed LLM routers schedule whole requests across sites: Topcu et al. [18] (-26% to -60% CO_2 via multi-DC MILP at $\approx 3\%$ SLO violation), SLIT [19] (carbon, water, TOU routing across 5 sites), CarbonRouter [20] (-16.1% to -23.1% operational CO_2 pricing multi-turn recomputation), and Bernhard and Yardimci [21] (2-DC live testbed plus CONUS MOER replay yielding 22.7%–51.0% cuts). Zero of these eight WAN schedulers (0/8) incorporate P/D pool separation, cross-region KV forwarding, embodied carbon, or serverless keep-alive.

3) L1.C Carbon-Aware Serverless Keep-Alive and Loaders ($n = 7$): Built on Azure Functions traces [36], CPU controllers EcoLife [37], CASA [38], LowCarb [39], and LACE-RL [14] (-51.7% cold starts, -77.08% idle carbon) adapt keep-alive windows, while GPU loaders ServerlessLLM [23] (10.3 s 70B load) and Festina [40] (-56% energy, -85% cold starts, plus -7% P/D) cut reload latency without testing keep-alive polarity across clean vs. coal grids.

4) L1.D Phase-Disaggregated and Engine-Level LLM Serving ($n = 17$): Local engines vLLM [29], Sarathi-Serve [6], Wilkins [31], ThrottLL'eM [44], SPROUT [16] ($> 40\%$ carbon cut via token LP), GreenCache [17] (5.6%–25.3% GPU vs. SSD total cut), ETCInfer [55] (33.1% energy cut via CRAC/DVFS without grid CI), DynamoLLM [7], TAPAS [45], and Fernandez [30] operate within single grids. Among the 8 disaggregated P/D designs (Splitwise [4], DistServe [5], Mooncake [28], DualScale [8], Li [32], Yan [9], Festina [40]), only GreenLLM [15] integrates grid CI and GPU lifetime reuse (-31.3% to -40.6% total carbon on one GCP cluster), while 0/8 span multiple regions, WAN KV gating, or serverless keep-alive.

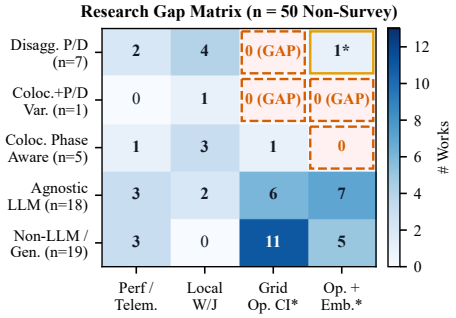


Fig. 3. Intersection of phase awareness and carbon/energy signals across $n = 50$ non-survey works. GreenLLM [15] populates P/D + Op.+Emb. (1⋆).

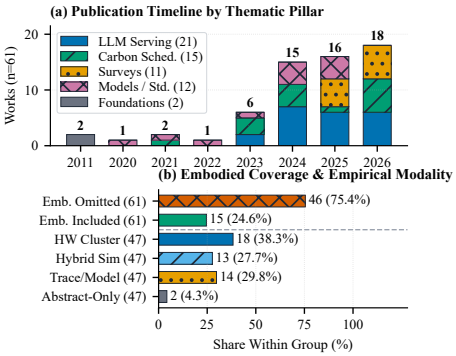


Fig. 4. Corpus bibliometrics from 2011 to Nov. 2026 ($n = 61$): (a) 2024–2026 growth (55.7% in 2025–2026); (b) 75.4% embodied omission ($n = 61$) and empirical split ($n = 47$).

5) L1.E Embodied Hardware and Lifecycle Models ($n = 11$): GHG studies [2], [3], [41], [42], LLMCarbon [25], Sinha [43], Hewage [46], EcoServe [24] (9 yr/3 yr lifetimes), Nguyen [50], HungryAI [53], and Vidur+Vessim [49] establish Scope 3 dominance on clean grids but omit phase-disaggregated control, WAN KV gating, and keep-alive.

6) L1.F Foundations and Traces ($n = 3$): NIST SP 800-145 [27], CloudSim [48], and the Azure LLM Inference Trace 2024 [26] supply taxonomy and trace baselines.

B. Gap Matrix and Closest Related Work Comparison

Cross-tabulating the 50 non-survey works in Fig. 3 shows that 46/61 works (75.4%) omit embodied accounting (Fig. 4b), while the 47 empirical works split across physical clusters (18/47, 38.3%), hybrid testbed+simulators (13/47, 27.7%), trace/analytical models (14/47, 29.8%), and abstract-only records (2/47, 4.3%). Table II maps the ten closest 2024–2026 systems and design agendas directly against this work across all four architectural axes, and Table III summarizes the broader L1.B–L1.F classes. Across all 61 works, **zero implemented systems** jointly combine P/D disaggregation, cross-region carbon routing with explicit WAN KV transfer gating, host plus GPU embodied carbon amortization, and serverless checkpoint keep-alive.

TABLE I
AUDIT OF PRIOR SURVEYS AND RESEARCH AGENDAS (L1.A, $n = 11$)

Work	Year	Venue / Status	Primary Scope	Grid CI	Emb.	LLM	P/D & KV	Verified Gap Summary
Hewage et al. [51]	2026	arXiv (28 pp.)	Latency-sensitive cloud	✓	✓	×	×	0 LLM/P/D/KV hits; cites $> 60\%$ CO_2 cut
Yang et al. [1]	2025	arXiv (35 pp.)	Container orchestration	✓	×	Min.	×	Covers SCI, serverless, GPU; 0 P/D/KV/emb.
Alam et al. [56]	2026	*Computing* (25 pp.)	Cloud carbon RL/heur.	6.3%	×	×	×	70% (44/63) sim-only; 0 GPU/LLM/emb.
Agarwal et al. [57]	2026	*ACM CSUR* (36 pp.)	Serverless adaptation	4/40	×	Min.	×	Sustainability secondary; 0 P/D/KV/emb.
Besozzi et al. [58]	2025	*IEEE Access* (46 pp.)	Serverless arch.	2 hits	×	×	×	29.5% GPU support; 0 LLM/P/D/emb.
Miao et al. [59]	2025	*ACM CSUR* (35 pp.)	Generative LLM serving	Min.	×	✓	✓	Covers KV & disagg; 0 emb./CI/water
Bai et al. [61]	2026	*ACM CSUR* (70 pp.)	LLM serving efficiency	✓	×	✓	×	PUE·kWh·CI metrics; 0 disagg/cold/emb.
Jiang et al. [60]	2026	*ACL Findings* (27 pp.)	KV compression	×	×	✓	✓	Recommends J/tok; 0 grid CI or emb.
Pan et al. [52]	2025	arXiv (25 pp.)	Data-centric LLM sys.	×	×	✓	✓	29 prefill, 12 disagg; 0 carbon/energy
Zhen et al. [54]	2025	arXiv (20 pp.)	Efficient LLM serving	Min.	×	✓	✓	6 P/D designs; brief Sec. 6.5 energy note
Sisodia [22]	2026	arXiv (13 pp.)	`llm-d` SICP agenda	Prop.	Prop.	✓	✓	Formalizes agenda Eq. (1)/(8); 0 expts.

TABLE II
CLOSEST RELATED WORK (2024–2026) VS. THIS WORK ACROSS ALL ARCHITECTURAL AXES

System / Work	Scope	P/D Split	Multi-DC	Grid CI	WAN KV Gate	Host+GPU Emb.	Serverless KA
GreenLLM [15]	Single GCP cluster	✓(het.)	×	✓	×	GPU only	×
Topcu et al. (Penn State) [18]	16 US/EU DCs	×	✓	✓	×	×	×
Sisodia (`llm-d` SICP) [22]	Research agenda	Agenda	Agenda	Agenda	Agenda	Agenda	×
CarbonRouter [20]	4 US DCs (multi-turn)	×	✓	✓	Recompute only	×	×
Cleanest Grid [21]	2-DC live + CONUS replay	×	✓	✓(MOER)	×	×	×
SLIT [19]	5 global DCs	×	✓	✓(+water)	×	×	×
GreenCache [17]	Single cluster + SSD	×	×	✓	×	SSD only	×
SPROUT [16]	Single A100 cluster	×	×	✓	×	✓(5 yr)	×
Vidur+Vessim [49]	Single-DC co-sim	×	×	✓	×	Formula only	×
EcoServe [24]	Single-DC lifecycle	×	×	✓	×	✓(9/3 yr)	×
This Work (`P4 + S3`)	**Multi-DC + Serverless Sim**	✓	✓	✓	✓	**✓(4yr & 9/3yr)**	✓

TABLE III
SYNTHESIS MATRIX ACROSS EMPIRICAL SCHEDULING, LLM SERVING, AND LIFECYCLE CATEGORIES (L1.B–L1.F)

Category / Representative Works	Class	Scope	Phase	Grid CI	Emb.	KV WAN	Keep-Alive	Primary Knobs / Metrics
Wait Awhile [10], CarbonScaler [11], EcoVisor [34], Clover [47]	L1.B	Single DC	Non-LLM	✓	×	×	×	Temporal shift, MIG, batteries
GreenCourier [33], Caribou [13], Sukprasert [12], Calavaro [35]†	L1.B	Multi-DC	Non-LLM	✓	×	×	×	Geo-routing, FaaS DAGs, LP
Topcu [18]‡, SLIT [19], CarbonRouter [20]‡, Cleanest Grid [21]‡	L1.B	Multi-DC	Coloc. LLM	✓	×	×	×	MILP, recompute cost, MOER
CASA [38], EcoLife [37], LowCarb [39]†, LACE-RL [14], ServerlessLLM [23], Festina [40]‡	L1.C	Serverl.	Mixed/P/D	Mixed	EcoLife	×	✓	Dynamic TTL, NVMe I/O, MPS
vLLM [29], Sarathi [6], DynamoLLM [7], GreenCache [17], SPROUT [16], ETCInfer [55]‡	L1.D	Engine/DC	Coloc. / Aware	Subset	Subset	×	×	DVFS, token LP, SSD cache
Splitwise [4], DistServe [5], Mooncake [28], DualScale [8]‡, Li [32]‡, Yan [9]‡	L1.D	Intra-DC	Disagg. P/D	×	×	Intra-DC	×	P/D pools, 50% cap, RDMA
GreenLLM [15]‡	L1.D	Single DC	Disagg. P/D	✓	GPU	Intra-DC	×	Het. A100+T4/V100, spec. dec.
ACT [41]†, LLMCarbon [25], EcoServe [24]‡, Nguyen [50], Vidur+Vessim [49]‡, HungryAI [53]‡	L1.E	Lifecycle	Agnostic	✓	Subset	×	×	SCI Eq. (1), 4R (9/3 yr), MFU
NIST [27], CloudSim [48]†, Azure Trace [26]	L1.F	Found.	Spec./Trace	×	×	×	×	Taxonomy, cloud sim, traces
This Work (`P4 Cluster + S3 Serverless`)	–	**Multi-DC**	**Disagg. P/D**	✓	✓	✓	✓	**Hybrid A–D, SCI, GapHazard**

†Abstract-only verified (5 works). ‡Preprint / accepted conference preprint (16 works across corpus).

C. Detailed Comparison Against Closest Systems

- **Topcu et al. (Penn State) [18]:** Their geo-distributed MILP router schedules continuous-batching LLMs across 16 US/EU DCs (queueing simulator within $< 5\%$ TPOT, 9.1% TTFT, $< 2.5\%$ energy), cutting operational CO_2 by 26%–60% near $\approx 3\%$ SLO violations. Because sites stay homogeneous colocated A100 pools, it omits P/D separation, WAN KV transfer, embodied carbon, and serverless keep-alive, all isolated here.
- **GreenLLM [15]:** GreenLLM disaggregates prefill/decode with speculative decoding on heterogeneous A100 and reused T4/V100 GPUs inside one GCP cluster, pairing grid CI with 5–7 yr GPU reuse for 31.3%–40.6% total carbon cuts at 90% SLO. Staying intra-DC, it omits WAN KV bottlenecks, host/chassis embodied dominance ($\geq 90\%$ clean grids), and serverless keep-alive.
- **Complementary Works:** CarbonRouter [20] prices prompt recomputation across four US DCs and defers WAN KV transfer, while Sisodia [22] sketches an intra-DC `llm-d` objective without experiments or serverless modeling.

V. SIMULATION ARCHITECTURE AND EXPERIMENTAL DESIGN

A. Parameterized Simulation Scope Statement

All empirical results in this study derive from a parameterized discrete-event simulation built in Python using SimPy 4.1.2; no telemetry was collected from physical multi-continent multi-GPU clusters. Consistent with the precedent noted by Alam et al. [56] (where 70% of cloud carbon-scheduling studies evaluate wide-area policies through simulation), discrete-event modeling supports reproducible, multi-seed ablations across heterogeneous electricity grids that an independent

researcher cannot instrument simultaneously. Model constants are separated into published hardware and serving baselines (Table IV) and explicit calibration assumptions (Table V), driven by public CC-BY-4.0 traces.

B. Phase Execution, Power, and Dispatch Formulation

For request i arriving at time t_i with prompt length $L_{\text{in}}(i)$ and output length $L_{\text{out}}(i)$, prefill duration on GPU tier g scales linearly above base kernel overhead $\tau_{P,0}(g)$:

$$T_{\text{exec},P}(i,g) = \tau_{P,0}(g) + \tau_{P,\text{tok}}(g) \cdot L_{\text{in}}(i), \tag{2}$$

calibrated to Splitwise Table IV uncontended Llama-70B anchors (84 ms conv and 95 ms code on H100; 155 ms and 185 ms on A100) [4]. During continuous-batching decode over active resident KV size K_{res} (GB) on G_r tensor-parallel GPUs, step time increases with HBM bandwidth contention:

$$t_{\text{step}}(g, K_{\text{res}}) = \tau_{D,0}(g)\left(1 + \alpha_{\text{mem}}(g)\frac{K_{\text{res}}}{G_r\,\text{BW}_{\text{HBM}}(g)}\right), \tag{3}$$

anchored to 28 ms on H100 and 42 ms on A100 [4], [7], with mild step inflation only if decode TDP cap $c_{\text{cap}} < 0.50$. Under colocation (`B1`), concurrent prefill slows active decode progress to factor $r_{\text{coloc}} = 0.20$ [4], [6], whereas disaggregation isolates decode iterations. End-to-end metrics aggregate cold-start delay, queueing, execution, KV transfer, and RTT:

$$\begin{aligned}\text{TTFT}_i &= T_{\text{cold},i} + T_{\text{q},P,i} + T_{\text{exec},P,i} + T_{\text{kv},i} + \text{RTT}_i,\\ \text{TPOT}_i &= \frac{T_{D,\text{exit},i} - T_{D,\text{enter},i}}{L_{\text{out}}(i)}.\end{aligned} \tag{4}$$

Instantaneous per-GPU power scales sublinearly with Model FLOPs Utilization (mfu) and active power cap ratio $c_{\text{cap}} \in (0,1]$ ($c_{\text{cap}} = 1.0$ on prefill; $c_{\text{cap}} = 0.55$ on decode [4]), plus GreenSKU Genoa host ancillary draw $P_{\text{host}}(\text{mfu})$:

$$\begin{aligned}P_{\text{gpu}}(\text{mfu}, c_{\text{cap}}) = P_{\text{idle}} + (c_{\text{cap}}P_{\text{TDP}} - P_{\text{idle}})\\ \times \left(\min\left(1.0, \frac{\text{mfu}}{\text{mfu}_{\text{sat}}}\right)\right)^{\gamma},\end{aligned} \tag{5}$$

using saturation threshold $\text{mfu}_{\text{sat}} = 0.45$ and calibrated sublinear exponent $\gamma = 0.70$.

For Llama-2-70B FP16 GQA-8 ($s_{\text{kv}} = 0.3125$ MB/tok), total KV payload $S_{\text{kv}}(i) = s_{\text{kv}}L_{\text{in}}(i)$ transfers over intra-DC NVLink/IB ($\delta_{\text{layer}} \in \{5, 8\}$ ms [4], $\text{BW}_{\text{local}} = 400$ Gbps) or cross-region WAN:

$$\begin{aligned}T_{\text{kv}}(r_P, r_D) &= \text{RTT}(r_P, r_D) + \frac{8S_{\text{kv}}}{B_{\text{wan}}},\\ E_{\text{kv}} &= P_{\text{net}}\frac{8S_{\text{kv}}}{3.6\times 10^6 B_{\text{wan}}}\,\text{PUE}.\end{aligned} \tag{6}$$

Cross-region pipelined disaggregation (Path C from r_P to $r_D \neq r_P$) is admissible only if both a latency recomputation crossover condition and an operational carbon crossover condition hold:

$$\begin{aligned}T_{\text{kv}}(r_P, r_D) + T_{\text{q},D}(r_D) < \min\big(T_{\text{q},D}(r_P),\\ T_{\text{recomp}}(L_{\text{in}}, r_D)\big),\end{aligned} \tag{7}$$

$$\begin{aligned}E_P I_{r_P}(t) + E_{\text{kv}}\frac{I_{r_P}(t) + I_{r_D}(t)}{2}\\ + E_D I_{r_D}(t) < \hat{C}_{\text{local}}(i).\end{aligned} \tag{8}$$

Across all feasible paths $p \in \{\text{A: Coloc., B: Intra P/D, C: Cross P/D, D: Remote Clean DC}\}$ meeting TTFT slack ($0.85 \cdot \text{TTFT}_{\text{SLO}}$), hybrid router `P4` selects:

$$\begin{aligned}p^* = \arg\min_{p\in\mathcal{F}(i,t)} \Big[w_c \hat{C}_{\text{op}}(p,t)\\ + 40\,w_q \max\big(0,\, \text{occ}_D(p) - 0.60\big)^2\Big],\end{aligned} \tag{9}$$

with $w_c = 1.0$, $w_q = 0.15$, and decode pool occupancy occ_D; when bursts exceed slack, `P4` falls back locally to Path A or Path B.

In Track B (serverless GPU serving), an online causal EWMA tracker (`OnlineGapHazardTracker`, `S3`) estimates discrete inter-arrival gap probabilities $\mathbf{p}$ across bins $[0\text{–}1, 1\text{–}5, 5\text{–}10, 10\text{–}30, 30\text{–}60, >\ 60]$ s and selects dynamic keep-alive timeout $k^* \in \{1, 5, 10, 30, 60\}$ s [14] at idle entry:

$$\begin{aligned}k^*(t) = \arg\min_{k\in\mathcal{K}} \Big[\mathbb{E}[\tau_{\text{idle}}(k)]\, r_{\text{idle}}(t)\\ + p_{\text{miss}}(k)\left(C_{\text{cold}}(t) + \frac{\beta_{\text{SLO}}\,\mathbb{I}_{\text{SLO}}}{\max(\widetilde{I}_r(t), 0.04)}\right)\Big],\end{aligned} \tag{10}$$

where $r_{\text{idle}}(t)$ is per-second warm-idle carbon rate ($\lambda_{\text{idle}} = 0.20$ baseline [14] or hardware idle ratio), $C_{\text{cold}}(t)$ is reload carbon during T_{cold}, and $\widetilde{I}_r(t)$ is normalized regional intensity.

Finally, operational emissions C_{op} integrate node and WAN power multiplied by PUE $= 1.12$ and regional grid CI $I_r(t)$, while amortized embodied emissions C_{emb} sum host and GPU footprints over lifetimes $\text{LT}_{\text{host}}, \text{LT}_{\text{gpu}}$ (4 yr uniform; sensitivity 9 yr host / 3 yr GPU [24]):

$$\begin{aligned}C_{\text{emb}} &= \sum_{m\in\mathcal{N}}\left(\frac{M_{\text{host},m}}{\text{LT}_{\text{host}}} + \frac{G_m M_{\text{gpu},m}}{\text{LT}_{\text{gpu}}}\right)T_{\text{sim}},\\ \text{SCI}_{\text{tot}} &= \frac{C_{\text{op}} + C_{\text{emb}}}{N_{\text{tok}}/1000}.\end{aligned} \tag{11}$$

C. Verified Baselines, Traces, and Calibrated Assumptions

Tables IV and V summarize verified profiles and calibrated assumptions. Multi-region cluster experiments (`E1`) span 3 zones (6 nodes, 24 H100s; 2 nodes/region): `South Scotland` (mean 9.75 gCO_2eq/kWh, 0–129), `London` (mean 163.32 gCO_2eq/kWh, 13–326), and `TEPCO` (mean 581.64 gCO_2eq/kWh, 302.3–671.5), from Oct. 2025 UK NESO 30-min traces (1,344 slots) and FY2024 Japan TSO traces (17,520 slots, converted from kgCO_2/kWh via $\times$1,000). Workloads use Azure LLM conversational (`conv`: 19,366 reqs, 5.53 RPS, CV $= 1.094$, mean 1,154.7 in / 211.1 out) and coding (`code`: 8,819 reqs, 2.57 RPS, bursty CV $= 13.152$, mean 2,047.8 in / 27.9 out) traces [26] under DynamoLLM bucket SLOs (Short: 250/100; Medium: 400/100; Long: 2,000/100 ms TTFT/TPOT) [7].

Cluster runs compare five routing policies: `B1` (carbon-agnostic colocated round-robin), `B2` (intra-region P/D, 55% decode cap), `P3` (unconditional cross-region P/D when RTT $\leq$ TTFT slack), `P4` (hybrid Eqs. (7) through (9)), and `P5` (oracle 30-min grid lookahead).

Serverless runs (`E2`) compare `S1` (0 s), `S2` (30 s), `S2b` (60 s), and `S3` (GapHazard Eq. (10)) across `TEPCO`, `North Scotland` (mean 35.65 gCO_2eq/kWh, std 82.50, 0–355), and `South Scotland` using ServerlessLLM's 10.3 s 70B loader.

TABLE IV
VERIFIED HARDWARE, EMBODIED, AND LOADER PARAMETERS

Component	Verified Value(s)	Source
H100 SXM5 (4/node)	TDP 700 W, Idle 60 W, HBM 3352 GB/s, 124.9 $kgCO_2e$	[4], [24]
A100 SXM4 (4/node)	TDP 400 W, Idle 100 W, HBM 2039 GB/s, 121.5 $kgCO_2e$	[4], [24]
L4 Decode (8/node)	TDP 72 W, Idle 14.4 W, 300 GB/s, 41.2 $kgCO_2e$	[24]
GreenSKU Genoa	Peak 562.4 W, Idle 164.6 W, 921.6 $kgCO_2e$	[24], [26]
Decode Cap & KV	$c_{\text{cap}} = 0.55$; $\delta_{\text{layer}} = 5/8$ ms; PUE 1.12	[4]
70B / 13B Loaders	SLLM 10.3 / 2.0 s; Safe 48.0 / 8.2 s; PyT 84.0 / 14.0 s	[23]
Amortization Life	Uniform 4 yr; Asymmetric 9 yr host / 3 yr GPU	[2], [24]

TABLE V
EXPLICIT SIMULATOR CALIBRATION ASSUMPTIONS (NON-MEASURED)

Parameter	Default / Range	Justification & Caveat
Synthetic WAN RTT	0.5 / 20 / 30 / 200 ms	Intra-DC / Intra-UK / Intra-JP / UK–JP delay
WAN KV BW	10 Gbps ({10, 25, 100})	Commodity Ethernet to DC WAN sweep
Sublinear MFU γ	0.70 ($\text{mfu}_{\text{sat}} = 0.45$)	Sublinear GPU load-power curve
Cold Power Frac.	0.65 P_{TDP} ([0.45, 0.85])	Deserialization + PCIe DMA warm-up
24h Extension	Empirical block tiling	Scales 58-min Azure blocks to 24h CI

All suites run across 10 seeds (100–109) and report Student-t 95% CIs ($t_{0.975,9} = 2.262$) and two-sided paired Wilcoxon p-values ($\alpha = 0.05$).

D. Simulator Validation Against Published Baselines

Prior to multi-region sweeps, 24 validation checks (`validate_simulator.py`; Supplement Table S2) anchor the simulator: (1) **Splitwise Table IV**: H100 disaggregated TTFT matches within 13.73%–14.24% (modeled 5–8 ms stall), A100 within 8.81%–9.50%, and H100 TPOT at 0.00% (conv, 28.0 ms) and -9.68% (code); (2) **Splitwise Figs. 8–9**: batch-16 decode step holds flat at 29.25 ms from 100% to 50% TDP (1,563.5 to 1,015.0 W) before +6.0% slowdown at 40% TDP (31.01 ms); (3) **ServerlessLLM Fig. 6a**: 0.00% error across all six 70B (10.3, 48.0, 84.0 s) and 13B (2.0, 8.2, 14.0 s) loaders; and (4) **Li et al.** [32]: `B2` vs. `B1` energy scales from -20.01% (0.25x RPS) to -30.40% (1.50x) as P95 TTFT rises +11.4% to +44.3% at 2.00x RPS.

E. 24-Hour Trace-Tiling Sensitivity Ablation

Because raw Azure traces span 58.36 min (`conv`) and 57.27 min (`code`), I tested 24h block-tiling sensitivity across four modes ($N = 10$ `conv` seeds; `run_tiling_ablation.py`, Table VI, Supplement Table S3). Relative `P4` SCI_{op} savings stay invariant within 0.12 pp (60.70%–60.82% vs. `B1`; 44.07%–44.33% vs. `B2`) and SLO violations hold within 0.817%–0.835%.

VI. SIMULATION RESULTS AND DISCUSSION

Unless noted otherwise, all inferential comparisons span $N = 10$ paired random seeds (100–109). Because $p = 2/2^{10} \approx 0.002$ (0.00195) is the minimum attainable two-sided exact paired Wilcoxon signed-rank p-value for $N = 10$ whenever all

TABLE VI
24H TRACE-TILING SENSITIVITY ON `CONV` ($N = 10$ SEEDS, MEAN)

Tiling Regime	P4 SCI_{op}	vs. `B1`	vs. `B2`	P4 SLO (%)
`orig_280s_repl` (12 blks)	0.03203	-60.70%	-44.07%	0.824%
`perm_280s_norepl` (12 blks)	0.03163	-60.76%	-44.21%	0.817%
`fine_90s_perm` (38 blks)	0.03166	-60.76%	-44.18%	0.820%
`coarse_900s_perm` (4 blks)	0.03138	-60.82%	-44.33%	0.835%

10 seed pairs agree in sign (rank-biserial correlation $|r_{\text{rb}}| = 1.000$), I report paired mean differences (Δ), Student-t 95% confidence intervals, rank-biserial r_{rb}, and paired Cohen's d_z alongside exact p-values (full paired statistics appear in the online supplement).

A. E1: Multi-Region Cluster Carbon–SLO Frontier

Table VII and Fig. 5 summarize `E1` across 1h and 24h horizons. On conversational traffic (`conv`), hybrid routing `P4` lowers SCI_{op} by 60.77% at 1h ($\Delta = -0.05124$ [-0.05473, -0.04775] g/1k tok, $r_{\text{rb}} = -1.000$, $d_z = -10.50$, $p = 0.002$) and 60.70% at 24h ($\Delta = -0.04948$ [-0.05220, -0.04676], $d_z = -13.01$, $p = 0.002$) relative to carbon-agnostic colocation `B1`, and by 43.95%–44.07% relative to intra-region phase disaggregation `B2` ($d_z \in [-19.99, -11.29]$, $p = 0.002$). Because fixed hardware amortization adds $\text{SCI}_{\text{emb}} \approx 0.0090$ gCO_2eq/1k tok across equal-capacity clusters, total SCI_{tot} drops 54.92% (1h) and 54.64% (24h) versus `B1` at 0.81%–0.82% SLO violations (TTFT P95 = 504.5–506.4 ms), well below the 5% boundary.

By contrast, unconditional spatial disaggregation `P3` reaches lower SCI_{tot} (0.03858 at 24h) at the cost of tail degradation, breaching 5% at 24h (5.90% violations, TTFT P95 = 1,190.4 ms). `P4` cuts `P3` SLO violations by 83.57% (1h, $\Delta = -4.12$ pp [-5.12, -3.11]) and 86.04% (24h, $\Delta = -5.08$ pp [-6.16, -4.00], $r_{\text{rb}} = -1.000$, $p = 0.002$) for a +6.48%–6.79% SCI_{tot} trade-off, supporting **H1** and **H2**.

Under bursty coding completions (`code`, CV = 13.152, mean 2,047.8-token prompts; Fig. 5b), `P4` cuts SCI_{op} by 28.51% versus `B1` ($\Delta = -0.01763$ [1h], -0.01685 [24h], $d_z \in [-13.14, -9.73]$, $p = 0.002$) and 23.45%–23.51% versus `B2` ($p = 0.002$). Unconditional `P3` saturates the WAN with 192.5–193.1 GB/1k-req KV transfers, driving SLO violations to 44.45%–44.57% (TTFT P95 = 30.7–39.3 s). To trace where spatial gains originate, `P4`'s gates prune cross-region Path C cleanly to **0.00%** at 10–25 Gbps (Section VI-C). Consequently, `P4`'s operational carbon reductions come primarily from **Path D** (whole-request remote clean-grid execution: 66.13%–66.27% on `conv`, 46.08%–46.74% on `code`) paired with local intra-DC fallbacks onto Path B (33.54%–39.81%) and Path A (0.20%–14.11%), cutting `P3`'s `code` SLO violations by 85.22% (1h, $\Delta = -37.88$ pp [-38.21, -37.55], $d_z = -82.83$, $p = 0.002$) and 86.01% (24h, $\Delta = -38.34$ pp [-38.57, -38.10], $d_z = -118.80$). Oracle 30-min grid lookahead `P5` matches `P4` within 0.02% on carbon and 0.25% on SLOs ($p \geq 0.160$).

B. E2 & SW4: Serverless Keep-Alive and Loader Sensitivity

Table VIII and Fig. 6 reveal an inverse regional trade-off for dynamic keep-alive (`S3`). On carbon-intensive `TEPCO` (581.6 gCO_2eq/kWh), 24h GapHazard `S3` achieves the lowest

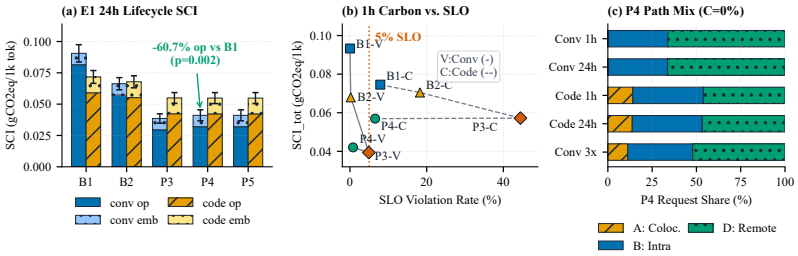


Fig. 5. Experiment `E1` cluster evaluation ($N = 10$ seeds, 95% Student-t CIs): (a) 24h operational plus embodied SCI; (b) 1h carbon–SLO frontier against the 5% boundary; (c) adaptive `P4` path mix showing Path C pruned to 0% at 10 Gbps WAN.

TABLE VII
EXPERIMENT `E1` MULTI-REGION CLUSTER METRICS (MEAN ± 95% CI, $N = 10$ SEEDS; PAIRED VS. `P4`)

Case	Policy	SCI_{op} (g/1k)	SCI_{tot} (g/1k)	TTFT P95 (ms)	SLO (%)	Effect (r_{rb}, p)	Verdict (H1/H2)
1h Conv	`B1` (Coloc.)	0.08431 ± 0.00681	0.09329 ± 0.00681	268.6 ± 0.0	0.0005 ± 0.001	$-1.00, 0.002$ (op)	Baseline
1h Conv	`B2` (Intra P/D)	0.05901 ± 0.00472	0.06799 ± 0.00472	300.3 ± 0.2	0.20 ± 0.04	$-1.00, 0.002$ (op)	Baseline
1h Conv	`P3` (Uncond.)	0.03040 ± 0.00306	0.03938 ± 0.00306	1062.3 ± 136.1	4.93 ± 1.02	$-1.00, 0.002$ (SLO)	Tail Degrad.
1h Conv	`P4` (Hybrid)	$\mathbf{0.03307 \pm 0.00365}$	$\mathbf{0.04205 \pm 0.00365}$	$\mathbf{506.4 \pm 2.1}$	$\mathbf{0.81 \pm 0.06}$	–	**SUPPORTED**
24h Conv	`B1` / `B2`	0.08150 / 0.05726	0.09055 / 0.06631	268.6 / 300.3	0.00 / 0.20	$-1.00, 0.002$ (op)	Baselines
24h Conv	`P3` (Uncond.)	0.02953 ± 0.00369	0.03858 ± 0.00371	1190.4 ± 156.0	5.90 ± 1.10	$-1.00, 0.002$ (SLO)	Breaches 5%
24h Conv	`P4` (Hybrid)	$\mathbf{0.03203 \pm 0.00434}$	$\mathbf{0.04108 \pm 0.00436}$	$\mathbf{504.5 \pm 3.7}$	$\mathbf{0.82 \pm 0.03}$	–	**SUPPORTED**
1h Code	`B1` / `B2`	0.06182 / 0.05773	0.07457 / 0.07048	592.8 / 3209.1	7.91 / 18.25	$-1.00, 0.002$ (op)	CV 13.15
1h Code	`P3` (Uncond.)	0.04441 ± 0.00385	0.05722 ± 0.00385	39317.8 ± 793.9	44.45 ± 0.28	$-1.00, 0.002$ (SLO)	WAN Collapse
1h Code	`P4` (Hybrid)	$\mathbf{0.04419 \pm 0.00388}$	$\mathbf{0.05694 \pm 0.00388}$	$\mathbf{1613.8 \pm 4.3}$	$\mathbf{6.57 \pm 0.25}$	–	**H1 Part./H2 Sup.**
24h Code	`B1/B2/P3`	0.05911/0.05525/0.04235	0.07171/0.06784/0.05495	612.5/3380/30725	8.46/19.47/44.57	$\leq -0.82, \leq 0.020$	Baselines
24h Code	`P4` / `P5`	**0.04226** / 0.04226	**0.05485** / 0.05485	**1598.9** / 1599.0	**6.24** / 6.23	$\lvert r_{rb}\rvert \leq 0.20, \geq 0.652$	**H1 Part./H2 Sup.**

TABLE VIII
EXPERIMENT `E2` 24H SERVERLESS KEEP-ALIVE METRICS ($N = 10$ SEEDS)

Region	Pol.	SCI_{op}	Emb. (%)	SLO (%)	Cold Starts
`TEPCO`	`S1` (0 s)	0.11773	22.2	40.25	6,271.6
`TEPCO`	`S2` (30 s)	0.11644	22.4	19.45	3,054.7
`TEPCO`	`S2b` (60 s)	0.12166	21.6	**13.03**	2,081.0
`TEPCO`	`S3` (GapHaz.)	**0.10825**	23.7	28.60	4,529.0
`N. Scot.`	`S1` (0 s)	0.00373	94.6	40.48	6,373.4
`N. Scot.`	`S2` (30 s)	0.00362	94.7	19.29	3,069.1
`N. Scot.`	`S2b` (60 s)	0.00378	94.6	**12.84**	2,073.4
`N. Scot.`	`S3` (GapHaz.)	**0.00323**	**95.0**	13.40	2,164.8
`S. Scot.`	`S1` (0 s)	0.00106	97.4	40.48	6,373.4
`S. Scot.`	`S2` (30 s)	0.00103	97.4	19.29	3,069.1
`S. Scot.`	`S2b` (60 s)	0.00108	97.4	**12.84**	2,073.4
`S. Scot.`	`S3` (GapHaz.)	**0.00094**	**97.6**	12.99	2,098.4

operational intensity ($SCI_{op} = 0.10825$), trimming operational carbon by 8.05% below scale-to-zero `S1` ($\Delta = -0.00948$ [-0.01058, -0.00838], $r_{rb} = -1.000$), 7.03% below 30 s static `S2` ($\Delta = -0.00819$ [-0.00873, -0.00764]), and 11.02% below 60 s `S2b` ($\Delta = -0.01340$, all $p = 0.002$). However, aggressive dirty-slot eviction leaves 28.60% SLO violations (4,529 cold starts) versus 19.45% (`S2`) and 13.03% (`S2b`), though well below `S1` (40.25%), partially supporting **H3**.

Conversely, on clean `South Scotland` (9.75 gCO_2eq/kWh), inverse intensity scaling in Eq. (10) extends warm retention during clean intervals, cutting cold-start SLO violations by **32.63%** relative to 30 s `S2` ($\Delta = -6.29$ pp [-6.60, -5.99], 12.99% vs. 19.29%, $r_{rb} = -1.000$, $p = 0.002$) while statistically tying 60 s `S2b` (12.84%, $p = 0.125$) at numerically lower SCI_{op} (0.00094, $p = 0.084$ vs. `S1`/`S2`). Embodied hardware accounts for 97.4%–97.6% of clean-grid lifecycle SCI ($SCI_{emb} = 0.03308$), supporting **H4**. Volatile `North Scotland` displays zero-carbon wind slots that erase operational differences ($p \geq 0.50$) while retaining the -30.50% SLO reduction versus `S2` ($\Delta = -5.88$ pp, $p = 0.002$).

In `SW4` (1h `TEPCO`, Fig. 6c), slow 70B loaders amplify `S3` operational carbon savings: -23.74% vs. `S1` under PyTorch (0.46686 → 0.35601 at 84 s) and -30.93% under Safetensors (0.35452 → 0.24488 at 48 s), whereas fast `opt_13b` (2.0 s) favors `S1` on carbon (0.00831 vs. 0.00936) at 16.02% vs. 3.60% SLO violations. Replacing the 20% idle ratio with H100's 8.6% hardware idle ratio (60 W/700 W) cuts `S3` SCI_{op} by another 14.43% (0.11193 → 0.09578).

C. Sensitivity Sweeps (`SW1`–`SW3`) and Clean Scale-Out Paradox (`H4`)

Fig. 7 and Table IX expose regime boundaries across load, WAN bandwidth, hardware tiering, and cluster capacity:

1) **SW1 Load Scaling (0.5x–3.0x conv):** At 3.0x load, `P3` saturates clean pools (71.94% SLO viol., TTFT P95 = 446.7 s), `B1` stalls via colocation (54.48% viol., TPOT P95 = 2,331 ms), and `B2` reaches 20.00%. Activating 11.30% Path A local spillover with Path B (36.62%) and Path D (52.08%), `P4` holds 2.89% violations ($SCI_{op} = 0.03483$ vs. `B1` 0.04943).
2) **SW2 Grid & Forecast Noise:** UK NESO forecasts, 30–60 min lags, and 2× noise shift `P4` SCI_{tot} by $< 0.3\%$. Across a low-differential UK pair (`South Scotland` + `North West England`), `P4` holds 0.08% violations ($SCI_{op} = 0.00513$) vs. `P3`'s 10.71% (0.00425).
3) **SW3 Hardware Tiers & WAN Path Mix:** Under `L4` decode nodes, `B2` (24.73%) and `P3` (53.47%) suffer decode SLO collapse, whereas `P4` holds 0.76% violations (Fig. 7b). Table IX explains WAN insensitivity: raising WAN bandwidth from 10 to 100 Gbps halves `P3`'s `code` violations from 44.45% to 20.78% (and high-rate 2.0x `conv` from 42.20% to 3.76%), whereas `P4` keeps Path C pruned

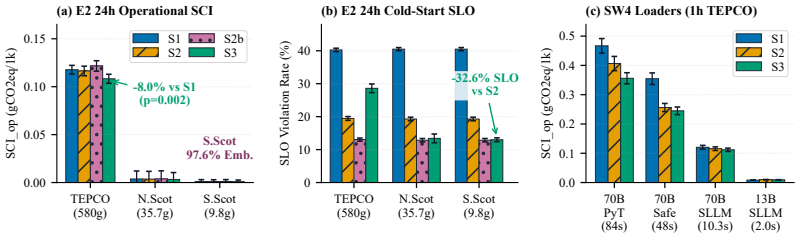


Fig. 6. Experiment `E2` and `SW4` serverless evaluation ($N = 10$ seeds, 95% CIs): (a) 24h operational SCI; (b) 24h cold-start SLO violations across grid intensities; (c) `SW4` loader sensitivity (84 s PyTorch down to 10.3 s ServerlessLLM and 2.0 s 13B).

TABLE IX
WAN BANDWIDTH SWEEP (`SW3c`) PATH MIX AND PERFORMANCE ($N = 10$ SEEDS)

Work.	BW	Pol.	SCI_{op}	SLO	P95 (ms)	A	B	C	D (%)
code	10G	P3	0.04441	44.45%	39,318	0.0	56.6	43.4	0.0
code	10G	P4	0.04419	**6.57%**	**1,614**	14.1	39.8	**0.0**	**46.1**
code	25G	P3	0.04403	32.12%	7,110	0.0	56.7	43.3	0.0
code	100G	P3	0.04391	20.78%	3,341	0.0	56.7	43.3	0.0
code	100G	P4	0.04419	**6.57%**	**1,614**	14.1	39.8	**0.0**	**46.1**
conv 2x	10G	P3	0.02481	42.20%	162,595	0.0	49.8	50.2	0.0
conv 2x	10G	P4	0.03191	**1.10%**	**881**	1.6	35.5	**0.0**	**63.0**
conv 2x	25G	P3	0.02502	9.63%	1,694	0.0	49.1	50.9	0.0
conv 2x	100G	P3	0.02488	3.76%	1,492	0.0	49.5	50.5	0.0
conv 2x	100G	P4	0.03193	**1.10%**	**881**	1.6	35.4	**0.02**	**63.0**

(**0.00%** at 10–25 Gbps; 0.02% at 100 Gbps `conv`) because clean-region whole-request routing (Path D: 46.08% `code`, 62.97% `conv`) and local disaggregation (Path B) dominate cross-region KV pipelining under 20–200 ms RTT. Under a strict 125 ms TTFT cap (`strict_code`), every fixed-capacity policy exceeds 5% (`P4` 48.69%, `B1` 53.30%).

4) **H4 Clean Scale-Out Paradox:** Across 24h `E1`, clean `South Scotland` reaches 93.0% embodied share on `code` (90.7% on `conv`). On all-UK nodes (where balanced `P4_equal` carries a 53.3% embodied share), adding four spare clean nodes (`P3_overprov_plus4`, 10 vs. 6 nodes) trims SCI_{op} by only -3.94% (0.00804 vs. 0.00837) while inflating net lifecycle SCI_{tot} by **+32.61%** under 4-year uniform amortization (0.02300 vs. 0.01735) and **+27.71%** under decoupled 9 yr host / 3 yr GPU lifetimes (0.01937 vs. 0.01516, Fig. 7c), supporting **H4**.

D. Summary Answers to Research Questions (`RQ1–RQ4`)

- **RQ1:** While GreenLLM [15] handles single-cluster P/D carbon and multi-DC routers [18]–[21] schedule colocated LLMs, 75.4% of 61 works omit embodied carbon, 0/8 geo-routers use P/D or WAN KV gating, and 0 implemented systems unify all four axes with keep-alive.
- **RQ2:** Hybrid routing (`P4`) cuts `conv` SCI_{op} by 60.70%–60.77% (SCI_{tot} 54.64%–54.92%, $|r_{rb}| = 1.000$) at $< 1\%$ SLO violation and trims `P3 code` SLO violations by 85.22%–86.01% by pruning Path C (0%) to rely on Path D and local fallbacks.
- **RQ3:** GapHazard (`S3`) saves 7.03%–11.02% `TEPCO` SCI_{op} (23.74%–30.93% for 48–84 s loaders) with 28.60% SLO violations, yet flips to cut clean `South Scotland` SLO violations by 32.63% vs. 30 s TTL.
- **RQ4:** Embodied carbon hits 93.0%–97.6% of clean-grid SCI, so four spare clean nodes raise net SCI_{tot} by +32.61% (4-yr) and +27.71% (9 yr/3 yr).

VII. OPEN CHALLENGES AND LIMITATIONS

A. Synthesized Open Research Challenges (`OC1–OC7`)

Integrating the 61-work literature synthesis (Section IV) with the simulation trade-offs in Section VI identifies seven open engineering challenges:

- **OC1 (Cross-Region KV-Cache Transport):** Because Eqs. (7) and (8) prune cross-region Path C for 99.98%–100% of requests even at 100 Gbps WAN, practical wide-area P/D splitting requires token-level KV compression [60] coupled with live grid carbon gates.
- **OC2 (Live Marginal vs. Average Carbon Control):** Most multi-region studies and the baseline `E1` sweeps rely on average regional CI; dispatching multi-megawatt bursts also requires marginal emission factors and curtailment signals.
- **OC3 (Sub-Minute Embodied Cluster Sizing):** Avoiding clean-grid over-provisioning penalties (+27.71% to +32.61% SCI_{tot} in `H4`) demands joint control loops coupling per-request routing with hourly host sleep/scale-down and extended GPU reuse [15], [24].
- **OC4 (Serverless Multi-Tenant Checkpoint Sharing):** Closing the 8.05% vs. 28.60% `TEPCO` carbon–SLO trade-off requires memory-deduplicated GPU weight caching across LoRA adapters and MPS slices [40].
- **OC5 (Water–Energy–Carbon Co-Optimization):** As SLIT [19] and CASA [38] demonstrate, routing exclusively toward dry solar/wind regions shifts cooling water stress unless water scarcity enters Eq. (1).
- **OC6 (Bursty Code-Completion Admission Control):** Under CV = 13.152, fixed 6-node clusters exceeded 5% SLO violations, pointing to elastic spot failover or prefill-priority admission control.
- **OC7 (Multi-Continent Disaggregated Hardware Testbeds):** While GreenLLM [15] evaluates single-cluster heterogeneous P/D on GCP and Bernhard and Yardimci [21] demonstrate live two-region colocated vLLM steering, no physical multi-continent testbed yet validates WAN P/D disaggregation, KV serialization, embodied accounting, and serverless keep-alive together.

B. Limitations and Threats to Validity

Six methodological boundaries govern the interpretation of this study:

1) **Trace-Driven Simulation Abstraction:** Quantitative policy outcomes derive from SimPy 4.1.2 discrete-event execution rather than instrumented multi-continent Kubernetes/vLLM clusters. Unmodeled RDMA tail jitter, PCIe thermals, and

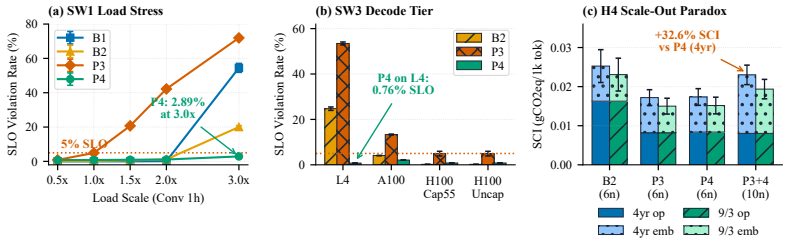


Fig. 7. Robustness and lifecycle evaluations ($N = 10$ seeds, 95% CIs): (a) `SW1` load scaling (`P4` holds 2.89% at 3.0x); (b) `SW3` decode GPU tiers (`P4` holds 0.76% on `L4`); (c) `H4` clean over-provisioning (+4 nodes) raising net SCI_{tot} by +27.71% to +32.61%.

host fragmentation may alter absolute latency margins, although relative bottleneck rankings match published hardware baselines (Section V-D).

2) **Calibrated Non-Measured Constants:** Synthetic WAN RTTs (20–200 ms), WAN bandwidths (10–100 Gbps), MFU exponent $\gamma = 0.70$, and cold-start GPU power fraction $f_{\text{cold}} = 0.65$ are literature-bounded engineering assumptions (Table V) swept across wide ranges in `SW3-SW4`.
3) **24-Hour Trace Tiling:** Raw Azure traces span 58.36 min (`conv`) and 57.27 min (`code`) [26]; Section V-E tests sensitivity across block lengths and replacement regimes.
4) **Static Dispatch vs. Dynamic Server Power-Off:** Inside Eq. (9), per-request `P4` scores operational carbon plus queue pressure; embodied amortization compares fixed versus over-provisioned cluster topologies (`H4`) rather than dynamically powering nodes down mid-run.
5) **Corpus Access and Preprints:** Five works (8.2%) were verified solely from abstracts [35], [39], [41], [43], [48], and 16 full-text entries (26.2%) are preprints (15 arXiv-only plus 1 accepted ATC 2026 preprint [40]) subject to peer-review revision.
6) **Search-Strategy Dependence and Screening Protocol:** Retained corpus counts depend on index cutoffs (November 2026) and query phrasing across Channels 1–3 (detailed in the online supplement). Screening and extraction were executed by a single author using generative AI search assistance rather than dual independent human reviewers, mitigated by inspecting every included work against primary full texts and Crossref DOIs.

VIII. Conclusion

This work investigated the joint intersection of carbon-aware cloud/serverless scheduling and phase-disaggregated LLM serving through a PRISMA-2020-guided review of 61 studies (56 full-text, 5 abstract-only) and a hardware-validated SimPy 4.1.2 simulation driven by Azure, UK NESO, and Japan TSO traces. While recent systems tackle individual dimensions such as single-cluster P/D carbon/GPU reuse [15] or colocated multi-region routing [18]–[21], 75.4% of reviewed studies omit manufacturing carbon, 0/8 cross-region carbon schedulers model P/D disaggregation or WAN KV transport, and no implemented system jointly integrates all four control axes alongside serverless checkpoint keep-alive.

Trace-driven multi-seed simulations show that unconditional spatial disaggregation (`P3`) suffers tail collapse under bursty coding traffic (44.45%–44.57% SLO violations) and 3× load spikes (71.94%). Hybrid carbon-and-queue routing (`P4`) preserves 0.81%–0.82% SLO violations on conversational workloads while cutting operational SCI by 60.70%–60.77% and lifecycle SCI by 54.64%–54.92% relative to colocation ($|r_{\text{rb}}| = 1.000$, $p = 0.002$), and lowers `code` SLO violations by 85.22%–86.01% below `P3` by pruning cross-region Path C (0%) in favor of Path D remote clean-grid execution and local Path A/B fallbacks. In serverless deployments, dynamic GapHazard keep-alive (`S3`) trims dirty-grid `TEPCO` operational SCI by 7.03%–11.02% (amplified to 23.74%–30.93% for slow 48–84 s loaders) while curbing clean-grid `South Scotland` SLO violations by 32.63% below static 30 s timeouts. Finally, hardware amortization constitutes 93.0%–97.6% of clean-grid lifecycle emissions, causing four redundant clean nodes to raise net total SCI by +27.71% to +32.61%. Future physical testbeds should pair power-capped intra-region disaggregation with WAN KV compression, sub-minute clean-capacity sizing, and shared NVMe checkpoint pools.

Declaration of Generative AI Use

Generative AI tools were used during manuscript preparation for literature search assistance, drafting, figure scripting, and language editing. Every reference, quantitative claim, simulator check, and figure was verified directly against primary sources and raw CSV outputs.

Data and Code Availability

External datasets (Azure LLM Inference Traces and Green-SKU tables, UK NESO Carbon Intensity API, and Japan FY2024 TSO factors) are publicly available; simulator code, PRISMA records, and reproduction scripts are hosted in a repository omitted here (*[anonymized for double-blind review]*).

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